Dear Editor,

I am pleased to submit for your consideration the manuscript entitled "Genomic Identity of the Sima de los Huesos Hominins: Convergent Evidence from Mitochondrial and Nuclear DNA", which I am submitting as an original research article to the American Journal of Biological Anthropology.

The Sima de los Huesos (SH) hominin assemblage (~430 ka, Atapuerca, Spain) represents the largest Middle Pleistocene fossil collection in the world, yet the apparent discordance between its mitochondrial and nuclear genomic signals has remained a persistent interpretive challenge in paleoanthropology since the pioneering sequencing work of Meyer et al. (2014, 2016). The present study resolves this discordance through a comprehensive reanalysis of both genomic compartments using current maximum likelihood phylogenetic methods with explicit outgroup rooting — an essential step that was absent in the original distance-based analyses.

The principal findings are as follows. Maximum likelihood reconstruction (IQ-TREE v3.0.1 and RAxML-NG v1.2.2, both with explicit Pan troglodytes rooting) consistently places Sima mtDNA as the basal sister lineage to the (Denisova + Neandertal) clade (bootstrap 93%; Robinson-Foulds distance = 0), not as sister to Denisovans as previously inferred from unrooted analyses. Nuclear D-statistics computed with ANGSD v0.941 on chromosome 1 (218,757,031 filtered sites) demonstrate unambiguous preferential affinity of the SH nuclear genome with Neandertals over Denisovans ($D = 0.7425$ vs. 0.6705; $Z = 30.6$; $p < 2 \times 10^{-16}$). Spatial interpolation of pairwise nuclear distances across Eurasia reveals a continuous geographic gradient with no detectable discontinuity between archaic taxa, indicating that geography is a stronger organising factor than time for Pleistocene nuclear genetic structure.

Together, these results demonstrate that the mito-nuclear discordance is not paradoxical but reflects incomplete lineage sorting in a transitional population: the SH maternal lineage predates the Neandertal–Denisova divergence, while the nuclear genome places this population firmly within the Neandertal stem lineage. These data are incompatible with assignment of the SH hominins to a generic Homo heidelbergensis and support recognition of Homo atapuercae (Bermúdez de Castro et al., 1997) as a distinct European taxon.

I believe this manuscript is well-suited for AJBA for several reasons. It directly addresses a long-standing interpretive problem in paleoanthropology using publicly available legacy data and transparent, reproducible methods. All genomic data are publicly archived (ENA: PRJEB10597, PRJEB6622; GenBank accessions provided). The findings are of broad relevance to the AJBA readership working in paleoanthropology, ancient genomics, and Middle Pleistocene human evolution — and carry particular significance given the site's location in Spain and the manuscript's inclusion of a full abstract in Spanish.

This manuscript has not been published previously and is not under consideration elsewhere. The author declares no competing interests. This research received no external funding.

I suggest the following researchers as potential reviewers, given their expertise in Middle Pleistocene genomics and paleoanthropology:
- Dr. Cesare de Filippo (Max Planck Institute for Evolutionary Anthropology) — ancient DNA, D-statistics
- Dr. Carles Lalueza-Fox (Institute of Evolutionary Biology, Barcelona) — Iberian paleoanthropology, ancient genomics
- Dr. Chris Stringer (Natural History Museum, London) — Middle Pleistocene hominin taxonomy

Thank you for your time and consideration.

Sincerely,

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